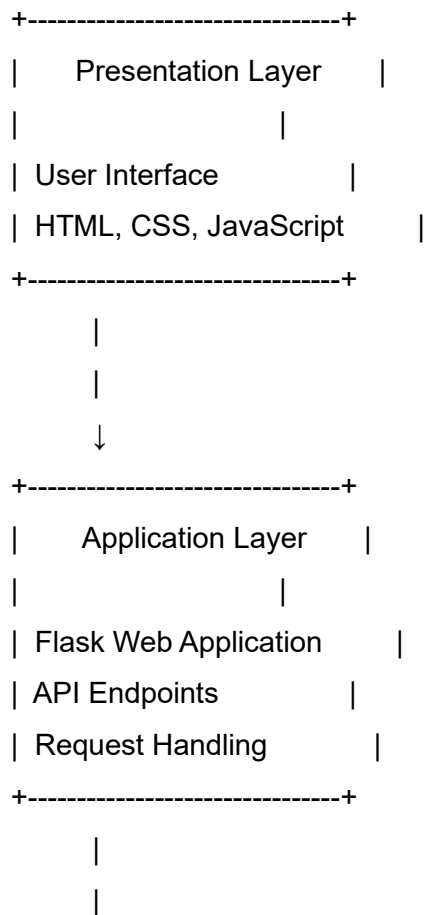


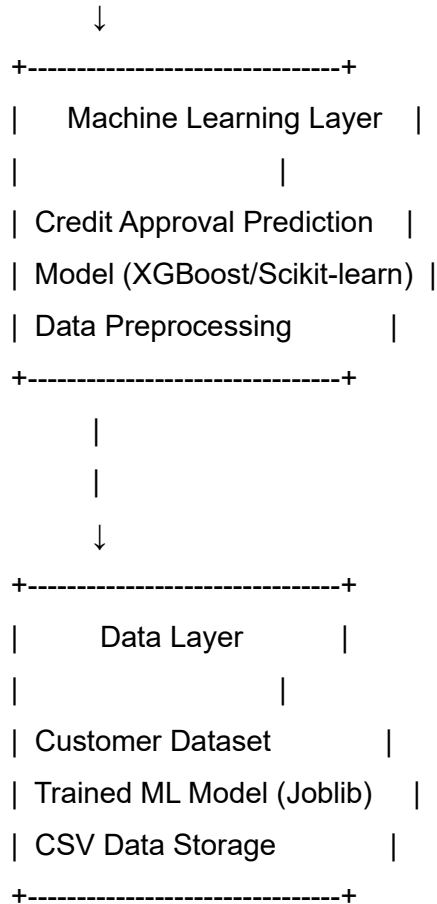
Solution Architecture

Date	15 July 2026
Team ID	
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.





Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides the user interface where customers enter their details and view credit card approval prediction results.	HTML, CSS, JavaScript
API Gateway / Backend API	Handles communication between frontend and machine learning model. Receives user inputs, processes requests, and returns prediction results.	Python Flask
Machine Learning Service	Performs data preprocessing, loads the trained model, analyzes customer attributes, and predicts approval probability.	Scikit-learn, XGBoost, Pandas, NumPy, Joblib
Database / Data Layer	Stores customer application data, historical records, datasets, and trained model files required for prediction.	CSV Dataset, Joblib Model Storage

Component Name	Description / Role in Architecture	Technologies Used
Deployment Layer	<i>Hosts the Flask application and provides reliable execution of the prediction service.</i>	<i>Gunicorn, Cloud Platform (Render/AWS)</i>